

Cybersecurity Training Diary

Day 10: Wireless Security

1. Topics Covered

✓ Wireless Encryption Protocols

Protocol	Security Level	Vulnerabilities	Best For
WEP	Broken	RC4 flaws (5 min crack)	Legacy devices
WPA	Weak	TKIP limitations	Temporary use
WPA2	Good	KRACK attack (patched)	Home networks
WPA3	Strong	Dragonblood (patched)	Enterprise

✓ Common Wi-Fi Attacks

```
graph LR
    A[Attacker] --> B[Evil Twin AP]
    A --> C[Deauth Attack]
    A --> D[WPS PIN Bruteforce]
    B --> E[Credential Harvesting]
    C --> F[Force Reconnection]
    D --> G[Network Access]
```

✓ Enterprise Wireless Security

- **802.1X Authentication** (RADIUS server)
- **EAP Methods:** PEAP, EAP-TLS
- **WPA3-Enterprise:** 192-bit encryption

2. Key Takeaways

WPA3-SAE prevents offline password cracking

MAC filtering is useless security theater

5GHz networks reduce wardriving risks

3. Practical Exercises

Exercise 1: Detect Rogue APs

```
# Use aircrack-ng suite
sudo airmon-ng start wlan0
sudo airodump-ng wlan0mon
```

Exercise 2: Secure Your Router

1. Disable WPS
2. Change default admin credentials
3. Enable WPA3 (or WPA2 AES-CCMP)

4. Questions for Reflection

Why do coffee shops still use open Wi-Fi instead of WPA3?

How could you securely configure IoT devices on Wi-Fi?

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